

|                                                         |                                                           |
|---------------------------------------------------------|-----------------------------------------------------------|
| Program : <b>Diploma in Instrumentation Engineering</b> |                                                           |
| Course Code : <b>4088</b>                               | Course Title : <b>Microcontroller and Interfacing Lab</b> |
| Semester : <b>4</b>                                     | Credits : <b>1.5</b>                                      |
| Course Category : <b>Program Core</b>                   |                                                           |
| Periods per Week: <b>3 (L:0 T:0 P:3)</b>                | Periods per Semester : <b>45</b>                          |

### Course Objectives:

- To train the student to the basic concepts and applications of Micro controller systems
- To give the student hands-on experience with the concepts of Micro controller systems

### Course Pre-requisites:

| Topic                    | Course code | Course name                  | Semester |
|--------------------------|-------------|------------------------------|----------|
| Digital fundamentals     |             | Digital circuits and systems | 3        |
| Electronics fundamentals |             | Applied physics 2            | 2        |

### Course Outcomes:

On completion of the course, the student will be able to:

| COn | Description                                                                                    | Duration (Hours) | Cognitive Level |
|-----|------------------------------------------------------------------------------------------------|------------------|-----------------|
| CO1 | Develop the assembly language program using 8051 kit                                           | 18               | Applying        |
| CO2 | Interface 8051 with ADC, DAC and displays                                                      | 8                | Applying        |
| CO3 | Write assembly language Programs for Timer, interrupt and serial communication using 8051 kit. | 8                | Applying        |

|     |                                        |   |          |
|-----|----------------------------------------|---|----------|
| CO4 | Interface 8051 with motors and sensors | 8 | Applying |
|     | Lab Exam                               | 3 |          |

### CO – PO Mapping:

| CO <sub>n</sub> | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 |
|-----------------|-----|-----|-----|-----|-----|-----|-----|
| CO1             | 3   |     |     |     |     |     |     |
| CO2             | 3   |     |     |     |     |     |     |
| CO3             | 3   |     |     |     |     |     |     |
| CO4             | 3   |     |     |     |     |     |     |

3-Strongly mapped

2-Moderately mapped

1-Weakly mapped

### Course Outline:

| Module Outcomes | Description                                                                       | Duration (Hours) | Cognitive Level |
|-----------------|-----------------------------------------------------------------------------------|------------------|-----------------|
| <b>CO1</b>      | <b>Develop the assembly language program using 8051 kit</b>                       |                  |                 |
| M1.01           | Familiarize Micro controller Kit                                                  | 3                | Understanding   |
| M1.02           | Write assembly language program on Arithmetic operation of hexadecimal numbers    | 3                | Applying        |
| M1.03           | Write assembly language program to Sort an array ascending and descending         | 6                | Applying        |
| M1.04           | Write assembly language program to Convert code such as Hexa to BCD, BCD to ASCII | 6                | Applying        |
| <b>CO2</b>      | <b>Interface 8051 with ADC , DAC and displays</b>                                 |                  |                 |
| M2.01           | Interface ADC                                                                     | 2                | Applying        |
| M2.02           | Interface Seven segment LED display                                               | 2                | Applying        |

|            |                                                                                                       |   |          |
|------------|-------------------------------------------------------------------------------------------------------|---|----------|
| M2.03      | Interface DAC                                                                                         | 2 | Applying |
| M2.04      | Interface LCD                                                                                         | 2 | Applying |
|            | Lab Test – I                                                                                          | 1 |          |
| <b>CO3</b> | <b>Write assembly language Programs for Timer, interrupt and serial communication using 8051 kit.</b> |   |          |
| M3.01      | Write assembly language program for Timer programming, Interrupt Programming.                         | 4 | Applying |
| M3.02      | Write assembly language program for Serial Communication                                              | 4 | Applying |
| <b>CO4</b> | <b>Interface 8051 with motors and sensors</b>                                                         |   |          |
| M4.01      | Interface DC motor                                                                                    | 2 | Applying |
| M4.02      | Interface Stepper motor                                                                               | 3 | Applying |
| M4.03      | Interface Temperature sensor LM 35                                                                    | 3 | Applying |
|            | Lab Test – II                                                                                         | 2 |          |

**Text / Reference**

| <b>T/R</b>     | <b>Book Title/Author</b>                                                                                                                                |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| R <sub>1</sub> | Kenneth J Ayala , The 8051 Microcontroller , - Thomson Third Edition Pearson                                                                            |
| R <sub>2</sub> | Subratha Ghoshal, 8051 microcontroller internals, instructions, programming and interfacing , Pearson                                                   |
| R <sub>3</sub> | Muhammad Ali Mazidi and Janice Gillispie Mazidi, The 8051 Microcontroller and Embedded Systems Using Assembly and C - Second Edition ,Pearson Education |

**Online Resources**

| <b>S.No</b> | <b>Website Link</b>                                                                                                                                                 |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1           | <a href="https://www.electronicshub.org/8051-microcontroller-introduction/">https://www.electronicshub.org/8051-microcontroller-introduction/</a>                   |
| 2           | <a href="https://www.electronicwings.com/8051/introduction-to-8051-controller">https://www.electronicwings.com/8051/introduction-to-8051-controller</a>             |
| 3           | <a href="https://nptel.ac.in/courses/Webcourse-contents/IIT-KANPUR/microcontrollers">https://nptel.ac.in/courses/Webcourse-contents/IIT-KANPUR/microcontrollers</a> |